SECRET-LEARN — SYSTEM ARCHITECTURE

Privacy-preserving ML · source: docs/secret_learn_architecture.drawio + src/secretlearn/ · github.com/chenxingqiang/Secret-Learn

APPLICATION & USE CASES

- **Healthcare** — hospital FL
- **Finance** — cross-bank fraud
- **IoT / Edge** — distributed sensors
- **Research** — multi-institution
- **Government** — census / tax

SKLEARN-COMPATIBLE API 191 algorithms

- **Linear 39** — Ridge Lasso SGD
- **Ensemble 18** — RF AdaBoost
- **Clustering 14** — KMeans DBSCAN
- **Decomposition 14** — PCA NMF ICA
- **SVM 7** — SVC SVR NuSVC
- **NN 4** — MLP BernoulliRBM
- **+95** — neighbors trees NB …

PRIVACY MODES src/secretlearn/{federated_learning,secret_sharing,split_learning}/

FEDERATED LEARNING
federated_learning/
FLLinearRegression
local PYU · HEU aggregate
191 algos · 3–5×

SECRET SHARING
secret_sharing/
SSPCA
SPU MPC ABY3/CHEETAH
191 algos · max privacy

SPLIT LEARNING
split_learning/
SLMLPClassifier
model split · HEU acts
191 algos · vertical FL

INTELLIGENT SYSTEM + JAX

- **algorithm_classifier.py** — src/secretlearn/algorithm_classifier.py
- **JIT / XLA** — xlearn backend
- **jax.vmap** — batch + autodiff
- **smart fallback** — original sklearn

SECRETFLOW DEVICES

- **PYU** — plaintext party compute
- **HEU** — homomorphic aggregate
- **SPU** — MPC ABY3 / CHEETAH
- **TEE** — trusted execution

INFRASTRUCTURE & HARDWARE

- **CPU 1.5–2.5×** — local JAX
- **GPU 5–8×** — CUDA
- **TPU 9–15×** — Google
- **M1/M2 2–4×** — Metal beta

Same sklearn call surface; mode directory picks FL / SS / SL · 191 × 3 = 573 implementations